



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Scalable and Reliable Services

SRS

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Modern services

Google

twitter



facebook

NETFLIX



Linked in

BitTorrent



Gmail
by Google

twitch

Apple iTunes

amazon

You Tube

skype

ebay

Pinterest



Common characteristics of modern services

A **service** is an activity or series of activities of more or less intangible nature that normally take place in interactions between a **service customer** and a **service provider**

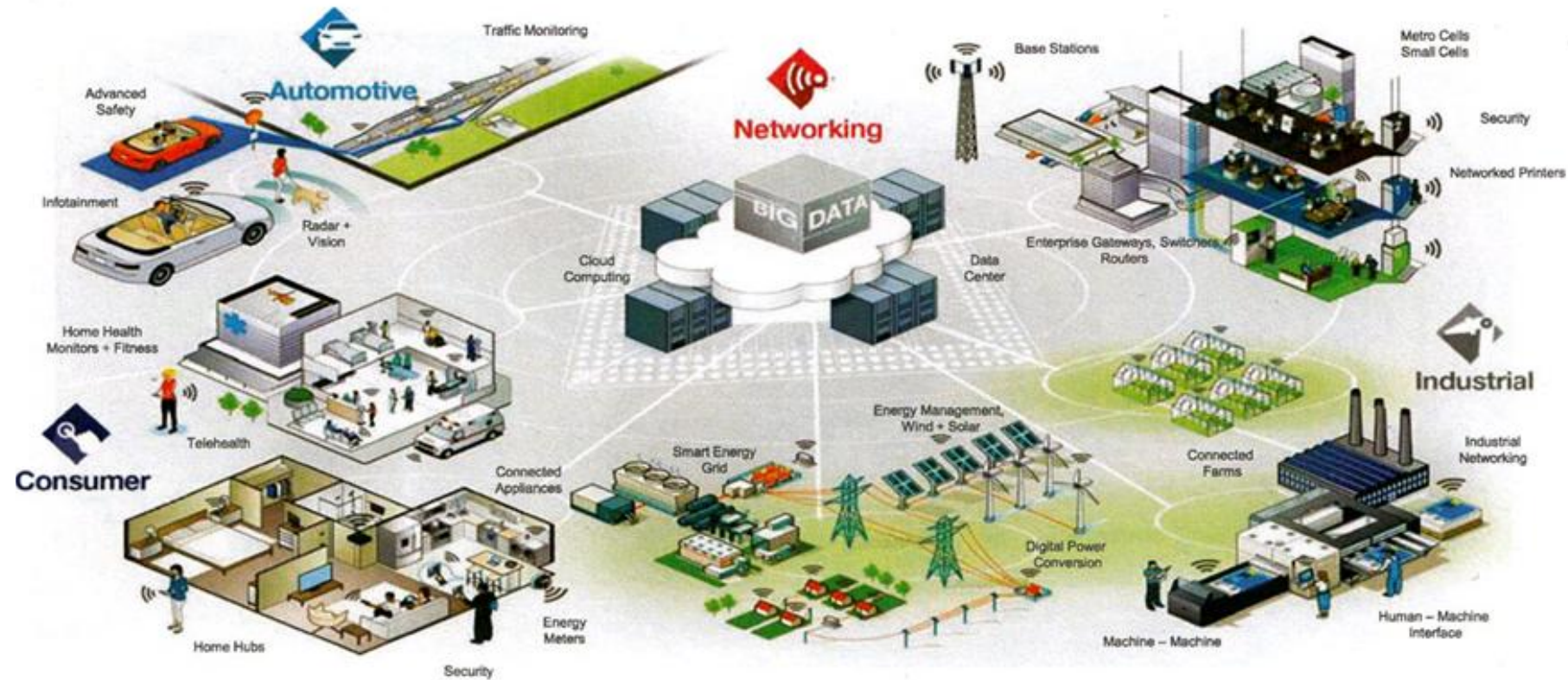
They work always: *resilient* and *secure*

They work well independently of the number of users: *scalable*

Their huge architectural complexity is *transparent*



We are building a Cyber-physical world



We are connecting people, data, things, machines, ..., that are enhanced by a lot of **sensors** and **actuators**

We are increasing **self-automation** of smart objects because ...

What's in a *service*?

“A service is the ability of a system (belonging to a *Service provider*) of delivering continuously one or more responses in the presence of specific requests to the system by a customer (*Service requestor*)”

1. There is a **customer** (hopefully, many)
2. There is a **service provider**
3. There is a **continuity** factor that the service provider must guarantee to the service requestor
4. There is a **quality** factor that is
 - stipulated through a formal contract → **Service Level Agreement (SLA)** including payments and penalties
 - implicit, but penalties are paid through loss of *reputation* and *trust*



From *technology delivery* towards *service delivery*

The alignment of IT, business methods and organizational practices influence innovation, productivity and client satisfaction in service industries

The fundamental principle behind present and future education models is that they should support the transition from a technology delivery approach towards a service delivery approach based on business goals



Different views

SERVICE

← **Customer's view**

**Design, Build, Test, Document, Monitor,
Fix, Improve ARCHITECTURES AND
APPLICATIONS**

← **The view in our course with some
AI support**



Scale → Complexity

**We have a problem of
scale of services**

**And *scale* changes (and
complicates) every well-
known problem in IT
research and industry**

**Billions of Users + Billions of “things”
*consuming***

**Tens of Millions of Services
*delivered by***

**Millions of Service Providers
*built on***

**Hundreds of Millions of Servers
*containing***

**Exabytes of Data
connected by
Terabytes Networks**

**Which (HW/SW) technology can
support these applications?**



Problem 1: Data management

Every two years we generate the same amount of information from the beginning of history to 2003

A week of the New York Times contains more information than a person of the 18th century was to come across in an entire lifetime

→ *Human minds and processors have reached their limits: steps ahead require collaboration*



Problem 2: Modern customers have zero tolerance for unresolved issues

- **Lost prospective customer**
 - If your competitor's service is responsive, you may have lost a sale
 - If the site did not work, or took too long, your customer may not return for a long time – if ever
- **Lost customer**
 - If this happens repeatedly, you have lost a customer
 - AND the customer may stop going to associated physical locations!
- **Lost reputation**
 - People talk about poor performance and reliability; word spreads especially through blogs and social networks
- **Lost lives**
 - In a cyberphysical world, safety is an important requirement



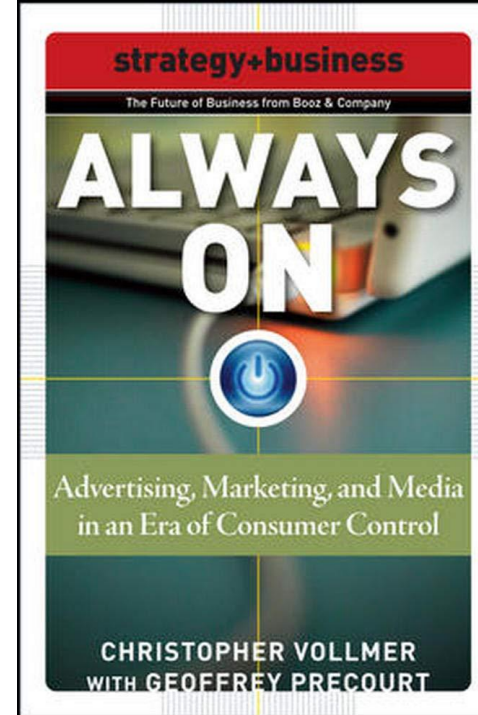
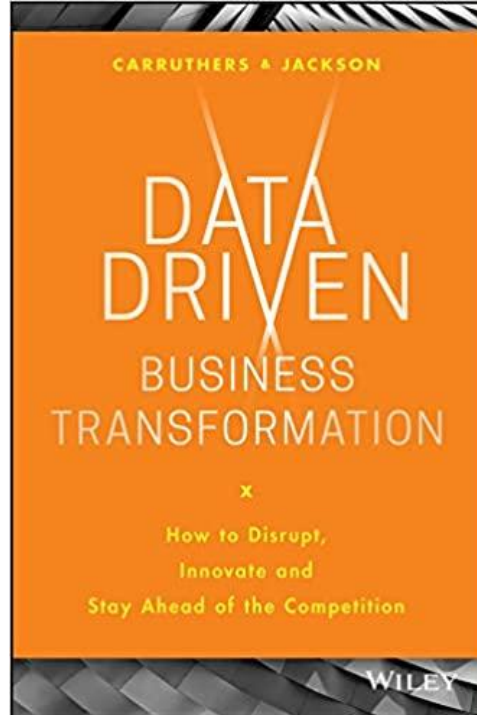
Problem 3: Today, speed, latency and resilience matter

There is substantial evidence suggesting that *performance* and *reliability* directly impacts revenue

- Google reported **20% revenue loss** due to a specific experiment that increased the time to display search results by as little as **500 msec**s
- Amazon reported a **1% sales decrease** for an additional delay of as little as **100 msec**s



Three requirements of *quality* of modern services



A novel cultural approach to the digital services

- **Reliability** is not only a collection of metrics or just a measure for the quality of your infrastructure and software
- It is a big-picture concept based on a collection of practices and cultural lessons that aim to improve service scalability and reliability by holistically incorporating
 - service architects
 - software experts developing and testing code
 - perspective of the users
 - inevitability of errors and failures



It is fundamental to **understand the service requirements** and the impact of reliability metrics, in order to:

- design and code with a focus on service level objectives
- evaluate and respect budgets
- improve incident response



Services and Business goals

It is fundamental to **understand the service requirements** and the impact of reliability metrics, in order to:

1. design and code with a focus on **service level objectives**
 2. evaluate and respect **budgets**
 3. improve **incident response**
- For each metric, organizations need to make decisions on **how much time loss and frequency of failures they can tolerated** without disrupting the system performance
 - They need to decide how much they can afford to spend on the service, infrastructure and support to meet certain standards of availability and reliability of the system
 - The resulting strategy is a **trade-off** between **cost** and **service levels** in context of the impact and requirements for maintaining a reliable and available service:
 - **Investments in critical infrastructures**
 - **Business value (*Return of investments*)**



Possible disturbances



INTENTIONAL ACTION

External attacks

Supply chain attacks

Internal attacks

NEGLIGENT ACTION

Human security errors

ERROR

INTERNAL or SUPPLIER HUMAN ERROR

System upgrade

Configuration error

Software fault

Poorly dimensioned

INTERNAL or SUPPLIER ACCIDENTAL EVENT

Failure (network, system)

Incident (black out, water leak, fire)



EXTERNAL ACCIDENTAL EVENT

Natural disasters

Changes in Law and Market

Technological innovations

Strikes, Riots

Diseases



Disturbances – *In this course*



INTENTIONAL ACTION

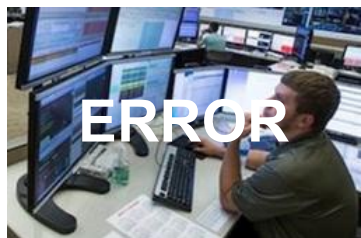
NEGLIGENT ACTION

External attacks

Human security errors

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INTERNAL or SUPPLIER
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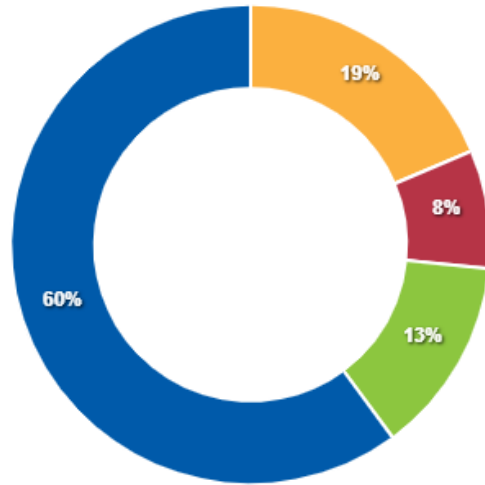
Strikes, Riots

Diseases



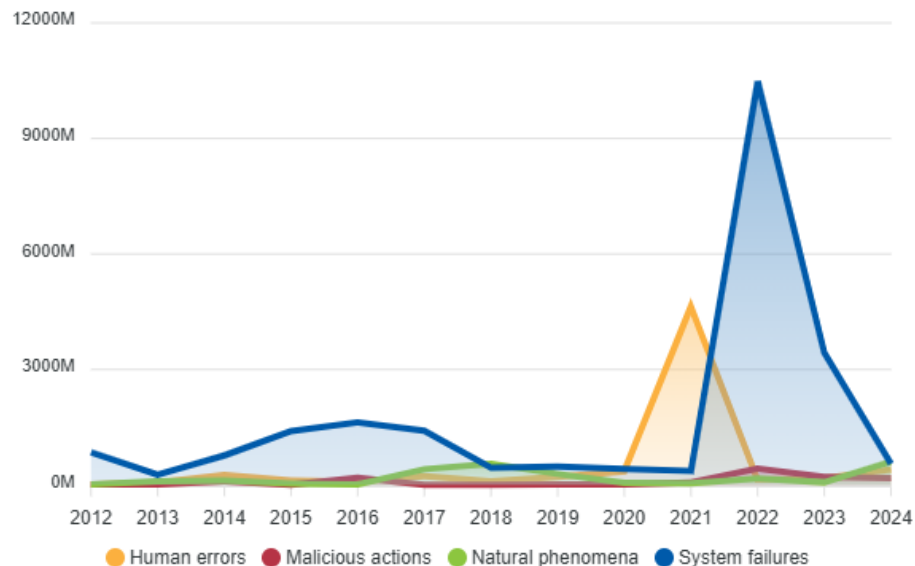
Telecom incidents

Nature of the incident

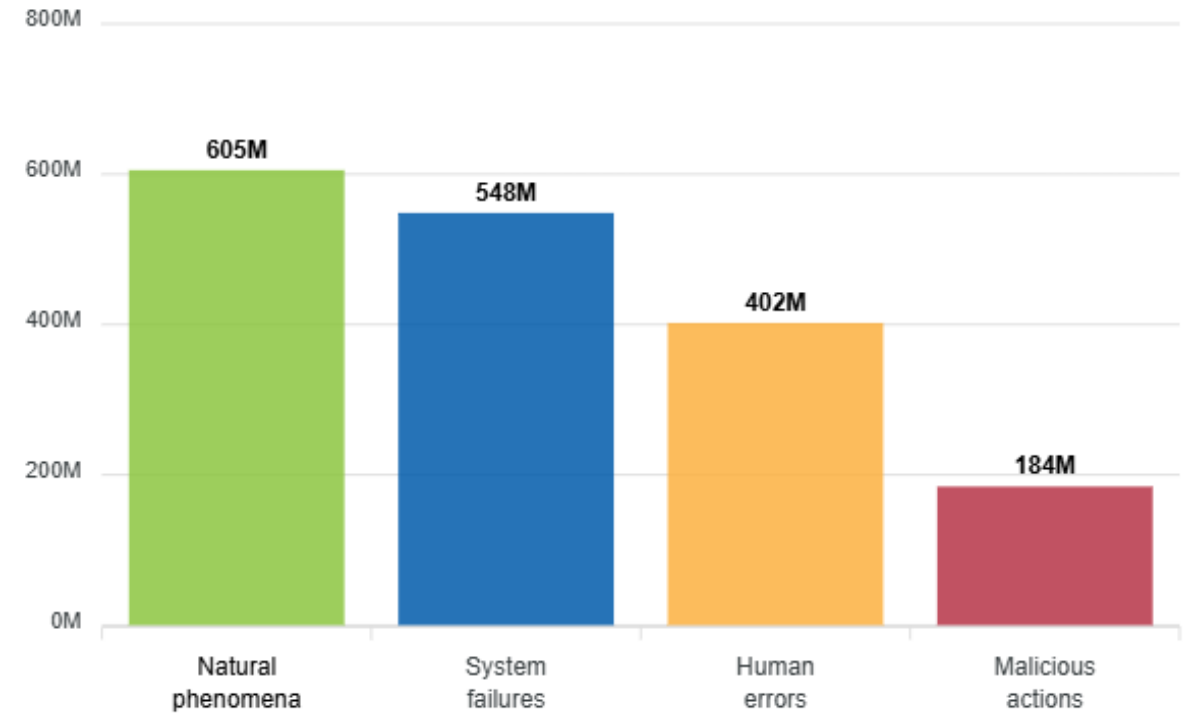


Human errors: 19% Malicious actions: 8% Natural phenomena: 13% System failures: 60%

User hours lost per root cause category



User hours lost per nature of incident



Availability and Reliability



Availability

Availability refers to the percentage of time that the system, infrastructure or service remains operational (**uptime**) in order to serve its intended purpose during the entire time interval of observation (**total elapsed time**)

$$\text{Availability} = \frac{\text{Total elapsed time} - \text{Downtime}}{\text{Total elapsed time}} = \frac{\text{Uptime}}{\text{Uptime} + \text{Downtime}}$$

Example

For a cloud-based service, that must operate 24/7, availability relates to the time that the data center is accessible and delivers the intended service as a proportion of the entire duration for which the cloud service is purchased.

Contract for one week 24/7 = 10,080 minutes – 4 hours of downtime

→ $164/168 = \mathbf{0.976}$ availability



Availability is typically measured on a '9' scale

*The famous
five nines*

Availability %	Downtime per year	Downtime per month	Downtime per week
90% (1 nine)	36.5 days	72 hours	16.8 hours
99% (2 nines)	3.65 days	7.20 hours	1.68 hours
99.5%	1.83 days	3.60 hours	50.4 minutes
99.9% (3 nines)	8.76 hours	43.8 minutes	10.1 minutes
99.95%	4.38 hours	21.56 minutes	5.04 minutes
99.99% (4 nines)	52.56 minutes	4.32 minutes	1.01 minutes
99.999% (5 nines)	5.26 minutes	25.9 seconds	6.05 seconds
99.9999% (6 nines)	31.5 seconds	2.59 seconds	0.605 seconds
99.99999% (7 nines)	3.15 seconds	0.259 seconds	0.0605 seconds

*Mission critical
services*



What percentage is acceptable?

90%

95%

99%

99.9%

99.99%

99.999%

99.9999%

Easy answer!
*The answer depends on
(how much critical is) the service*



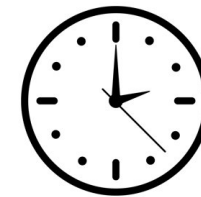
System view

Availability is also the probability that at time t the system works

Availability = 99% (two nines)

→ Downtime per year = $0,01 \cdot 525,600$ minutes
= 5256 minutes = 219 hours = approx. 9 days

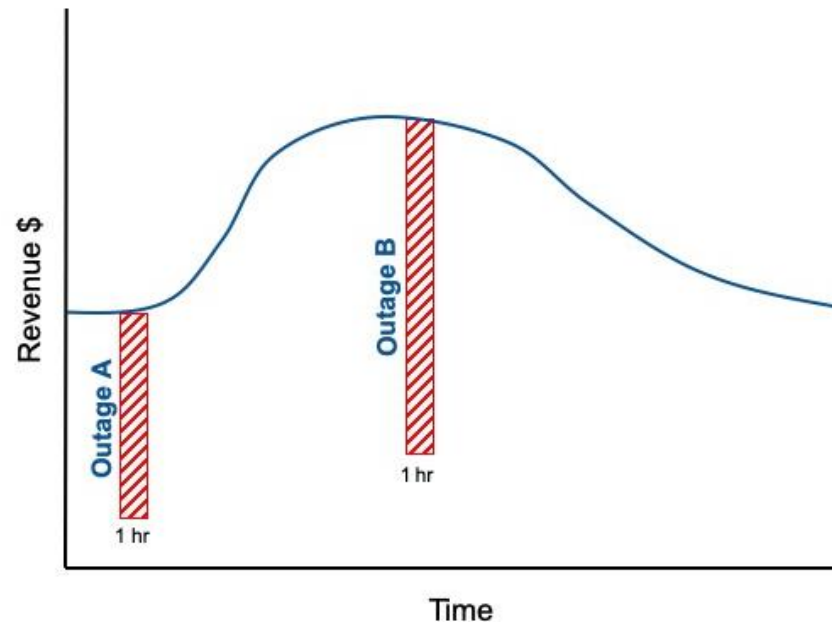
The system engineers of a provider typically evaluate the **quality of a service** in terms of **availability** that is measured as a percentage of the time that the service remains operational



Different times of outage, different impacts

However, units of time are not equal in terms of business impact: an interruption during the busiest part of the day would be much worse than an issue for late at night or early in the morning

Indeed, this “slow” period is often known as ***maintenance windows*** during which many companies schedule maintenance operations (e.g., patching, hardware replacement) because the impact of inoperability or limited operation is much smaller. If it is in the contract, it is acceptable; otherwise, it is a violation of the SLA



Reliability vs. Availability

Reliability refers to the probability that the system will meet certain performance standards in yielding correct output for a desired time duration

This may seem identical to how we define **Availability**, but there are some key differences

- 1) **Availability is a *system-oriented metrics***: it looks at whether the service is working, independently whether a user is accessing it or not
- 2) **Reliability is a *customer-oriented metrics***: it considers how a user perceives the *quality of a service*. If users accessed the service uniformly across all features and at all times, Availability would be equal to Reliability. **But this is generally never the case**



Reliability

Reliability is a **customer-oriented metrics**: it considers how a user perceives the *quality of a service*

- Reliability accounts for **workload** and **actual usage**. It converts availability (a *system metrics*) into a *customer measure*
- It can be also defined as the likelihood of a service functioning as expected when accessed by a user
 - Hence, mathematically, the Reliability is a subset of the Availability
- Reliability (and not Availability) should be considered as a basis for **Service Level Agreement (SLA)** by a customer

Not surprisingly, a **service provider** prefers *availability*, while **customers** should strive to get contracts based on *reliability* (e.g., an SLA for the day-weekdays; another for the night and weekend)



Example: *Which is the most reliable service?*

	Service A	Service B
User log-on page	97% availability	99% availability
Catalog search	97% availability	96% availability
Site settings page	97% availability	90% availability

- Service A would have the best availability and the best reliability if the workload was uniform
- On the other hand, Service B is better if
 - the log-on page is used by 100% of users
 - the catalog search is used by 90% of users,
 - the site settings page is used by only 30% of users

Reliability accounts for **workload** and actual usage



Service Level Agreement (SLA)

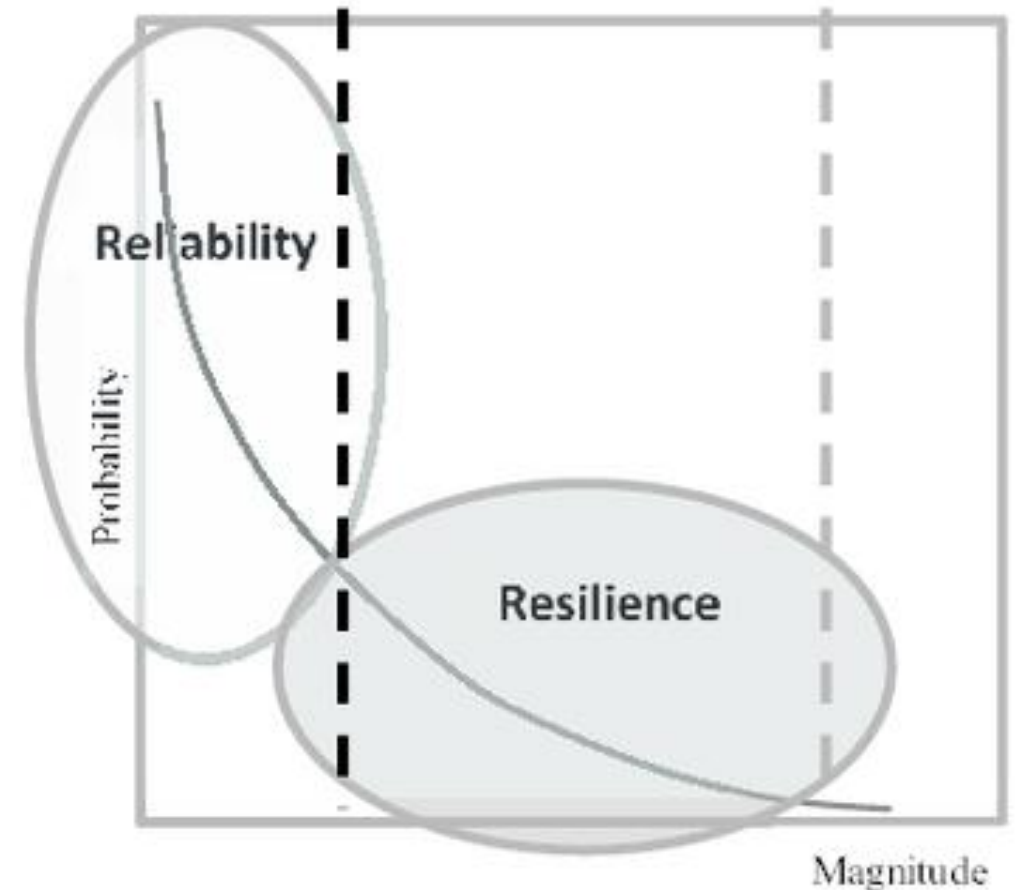
A service-level agreement (SLA) is a contract between a service provider and its customers. It documents what services the provider will furnish and defines the service standards the provider is obligated to meet

- When IT outsourcing emerged in the late 1980s, SLAs evolved as a mechanism to govern such relationships
- SLAs set the expectations for a service provider's performance and established penalties for missing the targets (and, in some cases, bonuses for exceeding them)
- The SLA should include not only a description of the services to be provided and their expected service levels, but also metrics by which the services are measured, the duties and responsibilities of each party, the remedies or penalties for violation, and a protocol for adding and removing metrics periodically

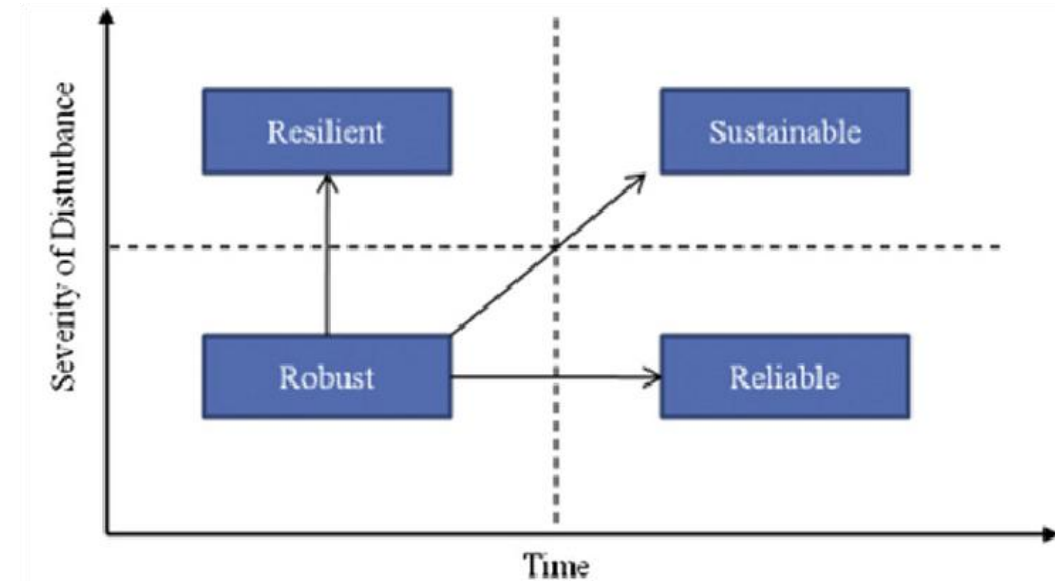
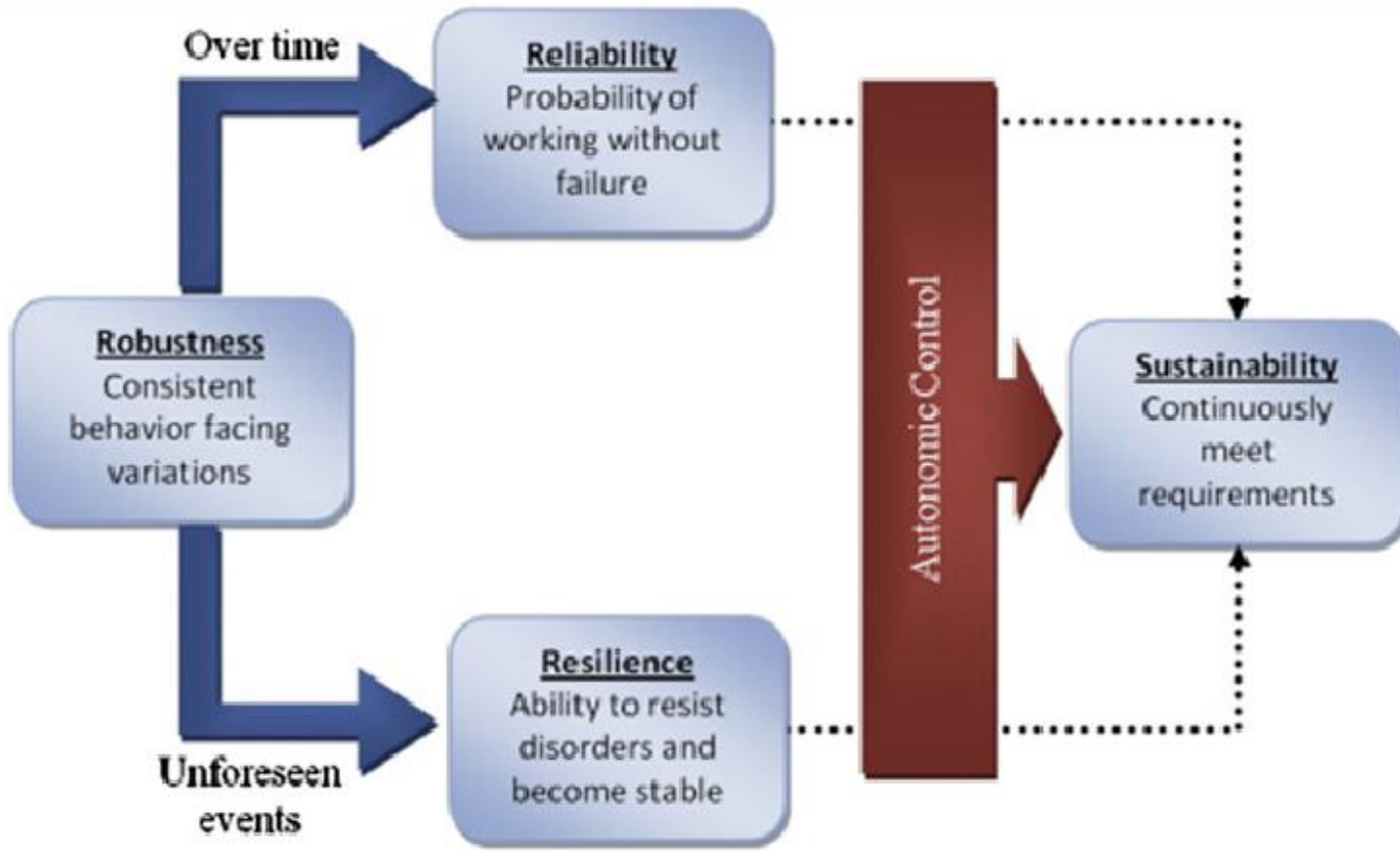


Reliability and Resilience

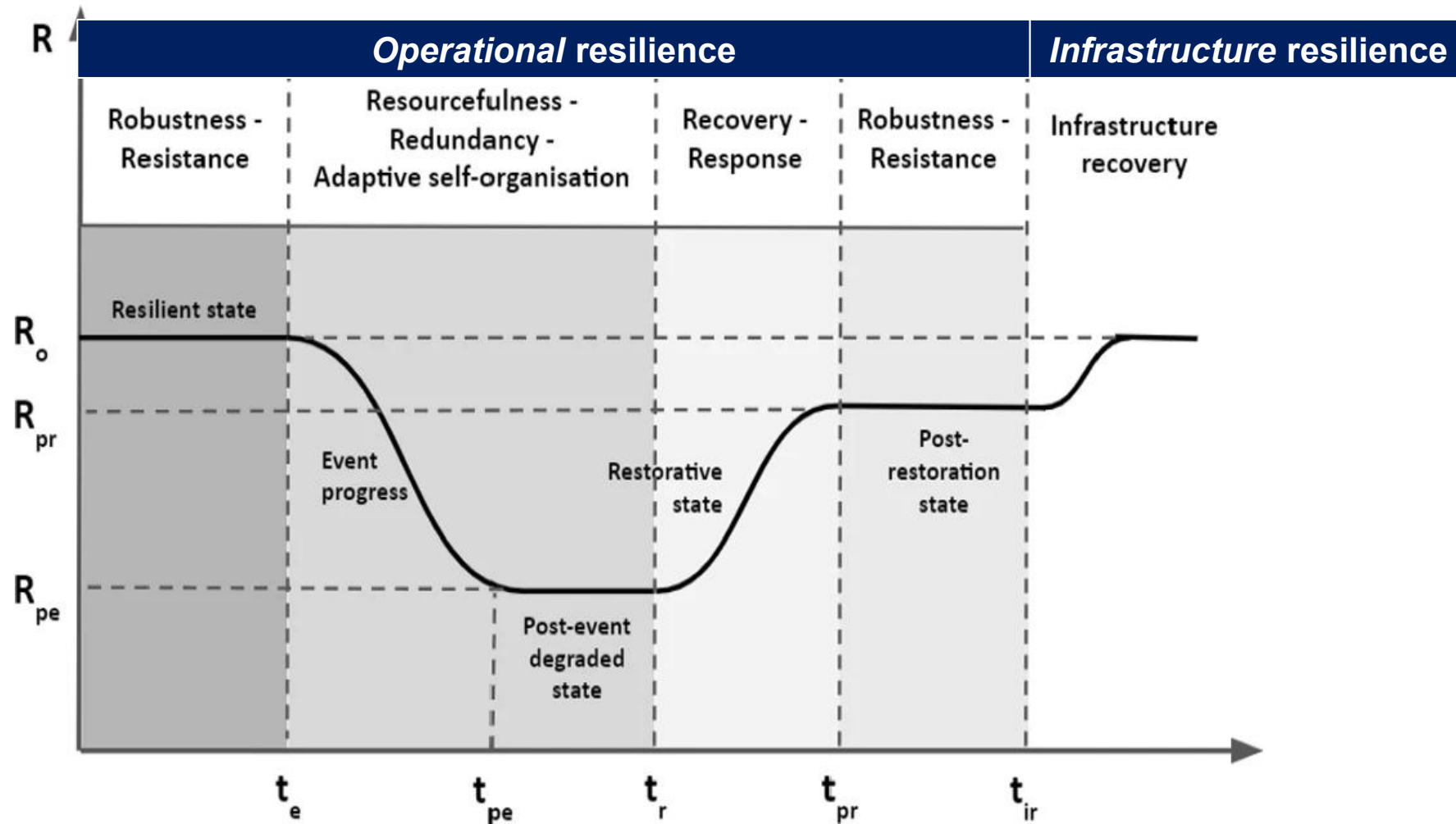
- **Reliability** is usually measured in relation to expected levels of service
- **Resilience** is the property and ability of the system to recover when the level of service has not been attained for whatever reason, but particularly when the system is subject to unexpected exceptional disturbances



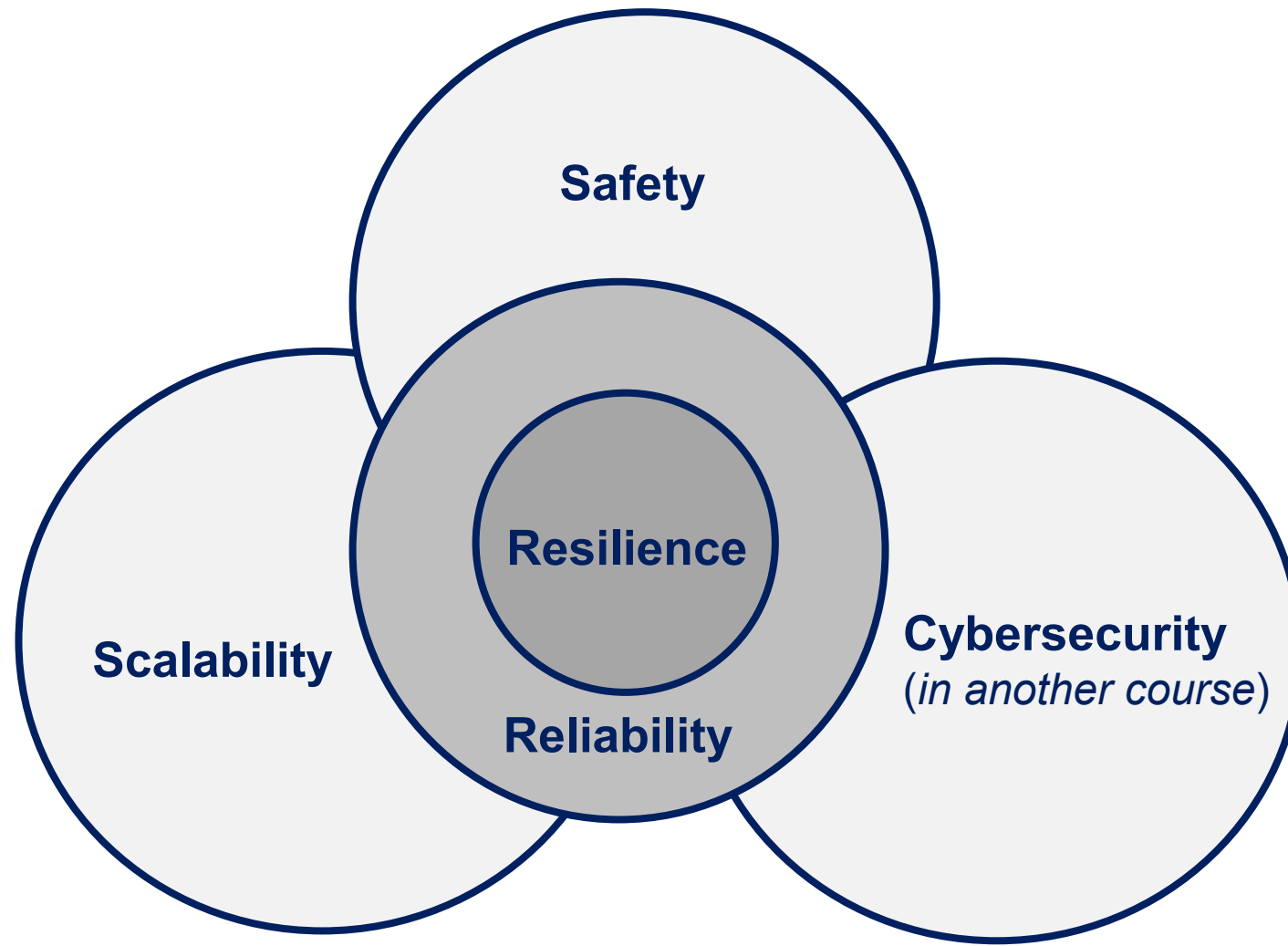
Robustness and Sustainability



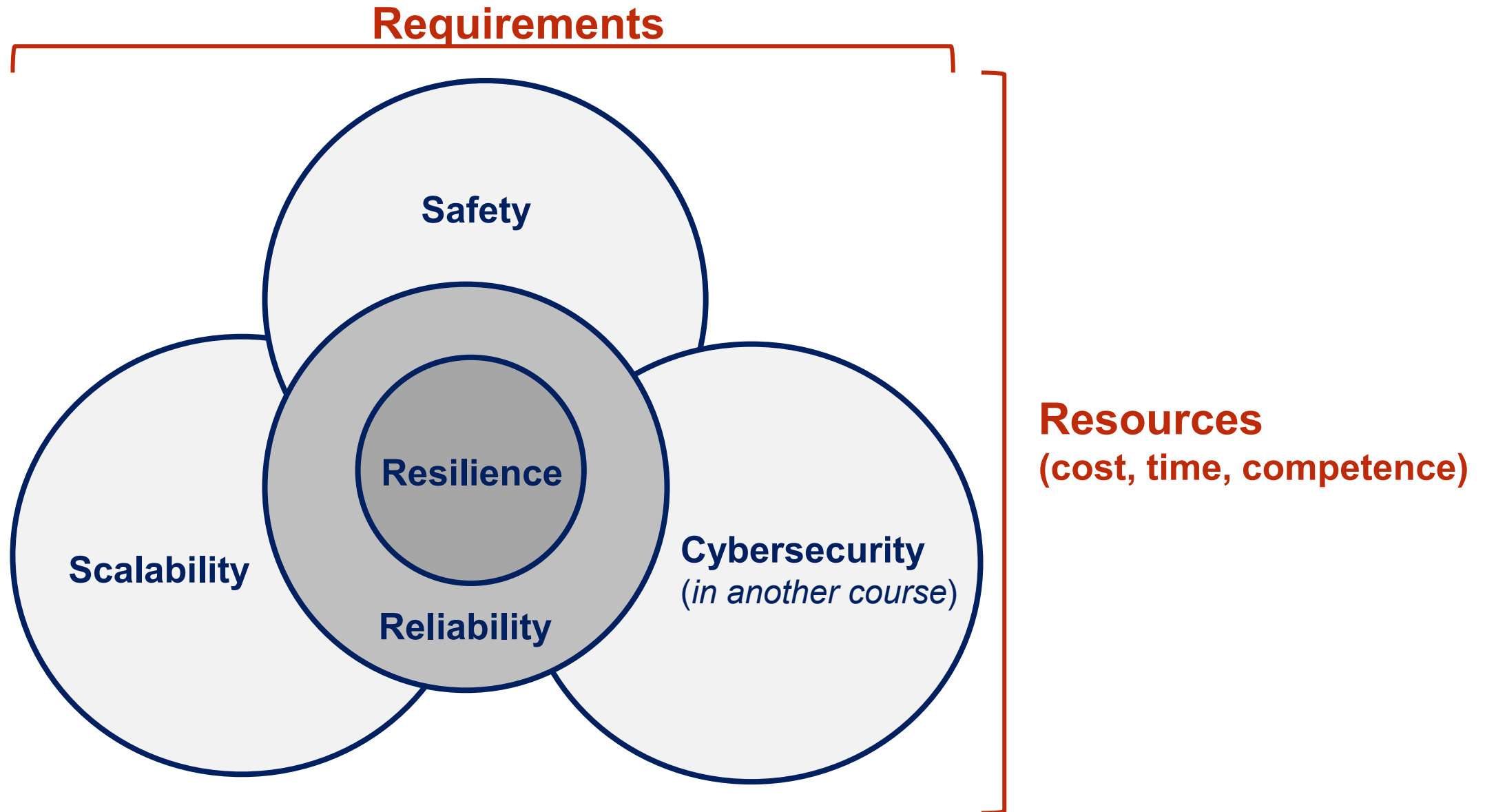
The overall vision



Design principles of modern services



Modern service (requirements) and constraints of the reality



Business cases with a strong commitment on reliability

Datacenters are so crucial to a digital services that the companies have determined exactly how much they lose whenever there is an outage

Mazen Rawashdeh (eBay): “**Depending on what goes down and when, the impact on the site is from \$2,000 to \$4,000 per second” (120,000 – 240,000€ per minute)**

For Amazon is about 300,000\$ per minute

For Facebook is about 150,000\$ per minute

What about your services?



Nationwide building society falls victim to IT outage

Nationwide joins McDonald's, Greggs, Sainsbury's and Tesco as the fifth major high street name to experience an IT service outage in the space of a week

By Alex Scroxton, Security Editor

Published: 22 Mar 2024 16:08

These outages appear to have occurred after **system upgrades** ran into problems. None of them have been attributed to malicious cyber activity.

Nationwide had reached a formal agreement to acquire *Virgin Money Bank* in a **£2.9bn deal** on March 21st 2025.

Many English questioned the logic of Nationwide making a multibillion-pound acquisition if it can't keep its IT running...



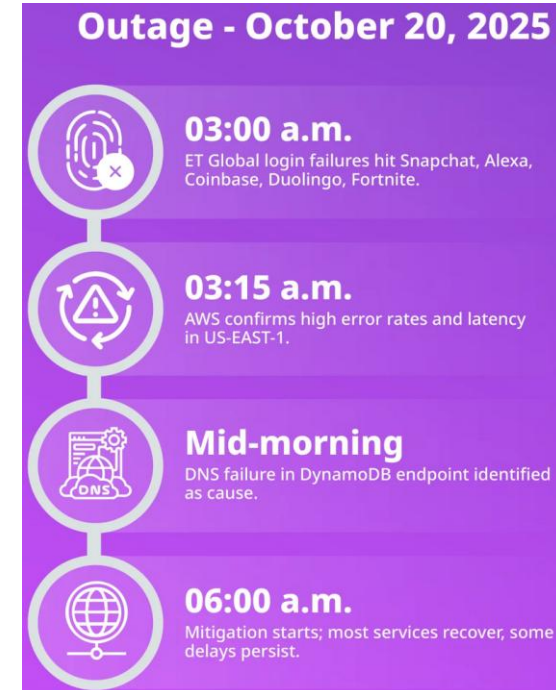
AWS outage – October 20, 2025

“The AWS outage of October 2025 didn’t just take down servers; **it silenced half the internet.**

For hours, major platforms like Snapchat, Alexa, and Coinbase went dark, sending millions of users into chaos and reminding businesses everywhere that **even the world’s biggest cloud can crash.**

What began as a **DNS issue** in the AWS US-EAST-1 region, specifically affecting the DynamoDB API endpoint causing that requests were not reaching their destinations because the DNS responsible for translating domain names into server addresses had stopped responding. This problem quickly spiraled into a major cloud outage, exposing just how dependent global operations have become on Amazon’s infrastructure.

For startups, enterprises, and digital agencies alike, the AWS downtime **revealed a fragile truth about modern cloud computing: convenience often comes at the cost of control.**”



Key 2025 Cloudflare incidents

- **March 21, 2025:** A R2 storage incident caused widespread write failure
- **July 14, 2025:** A 62-minute outage of the 1.1.1.1 DNS resolver.
- **August 21, 2025:** A network congestion incident prompted plans for a new, enhanced traffic management system.
- **September 12, 2025:** Dashboard and API outage caused by a recursive bug in a React hook, leading to excessive API calls.
- **November 18, 2025:** A significant ~6-hour disruption occurred, initially mistaken for a DDoS attack, but **actually caused by a configuration issue** with internal bot management software.
- **December 5, 2025:** A ~25-minute outage (starting ~8:47 UTC) occurred due to configuration changes meant to mitigate a vulnerability, resulting in 500-level errors.

CLOUDFLARE OUTAGE 2025
HOW ONE CONFIG FILE
CRASHED 20%
OF THE INTERNET



Learned lesson #1

Perfect solution → Unlimited budget

**Engineering is not about perfect solutions.
It's about doing the best you can with limited resources.
[Randy Pausch]**



Best solution to address *scalable* and *reliability* issues?

Replication, Replication, Replication

Replication of server machines

Replication of interconnections

Replication of data

Replication of application processes

Replication of people

...

They must be designed, implemented, managed



Any scalable and reliable architecture must avoid

System bottlenecks



Single points of failure



Another value of resilience?

Ability to adapt

The ***Choluteca bridge*** case, Honduras, beginning 1998

Built by the Japanese society Hazama Ando Corporation specialized in constructions resistant to earthquakes and cataclysms of all kinds (including Honduras hurricanes)



Expensive, solid, capable of surviving hurricanes, but *useless because unsuitable to face novel situations*

Physical components are *rigid*. Our software can be *adaptable*

